

Akhilan N. Ganesh

US Citizen | San Jose, CA | aganesh.work@gmail.com | (408) 250-3866

Skilled software engineer with an interest in backend applications and cloud development.

Experience

PerformanceStar LLC (Concept Tapisstry Inc.)

Jan 2024 – Mar 2025

Software Engineer

- Optimized apriori algorithm for XAI explanations (with a resulting speedup of 10-40x)
 - Developed and improved logic for generation of chemical vapor deposition recipes
-

Education

Bachelor of Science degree, Computer Science, **University of California-San Diego**, 3.89 GPA

June 2023

Projects

Snek Compiler (Rust)

Mar 2023 – Jun 2023

- Architected a compiler for the dynamically-typed CSE 131 Snek language
- Built compilation logic for functions, safe tail calls, heap allocation, and mark-compact garbage collection

Yodie ISA and Microprocessor (SystemVerilog, Python)

Jan 2023 – Mar 2023

- Led a team to design an ISA/microprocessor to handle error detection/correction & pattern match
- Optimized instruction set (to 9 bit instructions) & number of assembly instructions (cut by 50%)

Raytracer (C++)

Jan 2022 – Mar 2022

- Engineered a custom algorithm with ray casting, preliminary shading & recursive ray tracing
- Introduced a bounding box acceleration structure that improved performance logarithmically

Bujo: Web-based Bullet Journal (JS)

Apr 2021 – Jun 2021

- Collaborated in an 8-person team to build a bullet journal web app using industry Agile practices like Scrum
-

Skills

Programming Languages:

Java, C, C++, C#, C-CUDA, Golang,
Python, SWI-Prolog, Rust, SQL, SystemVerilog,
ARMv8, x86, HTML, CSS, JavaScript

Tools and Technologies:

Git/Github, Docker, SQLite, MySQL, PostgreSQL,
MongoDB, Node.js, Express.js, React, Next.js,
TailwindCSS, Pytorch, Quartus, QuestaSim, Linux

Undergraduate Courses of Interest

- Computer & Communication Networks (CSE 222A)
- Machine Learning (CSE 151A)
- Deep Learning (CSE 151B)
- Compiler Construction (CSE 131)
- Database System Implementation (CSE 132C)
- Principles of Computer OSes (CSE 120)

Website: <https://akhilanganesh.vercel.app>

LinkedIn: <https://www.linkedin.com/in/akhilan-ganesh/>

Github: <https://github.com/akhilanganesh>